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CURRICULUM VITAE

Leroy L. Jia

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Citizenship: U.S.

Positions Held

Instructor of Physics, Mathematics, and Engineering, Momentum Scholars	2026–Now
Staff Mathematician, Applied and Computational Mathematics Division, NIST	2023–2026
Instructor of Engineering, North Carolina School of Science and Mathematics	2023
Flatiron Research Fellow, Center for Computational Biology, Flatiron Institute	2018–2022
Supervisor: Prof. Michael Shelley	

Visiting Positions

Visitor, Courant Institute of Mathematical Sciences, New York University	2019–2020
Visiting Scientist, Department of Physics, Brown University	2018–2019

Education

Ph.D. Applied Mathematics, Brown University	2018
Advisors: Profs. Thomas Powers & Robert Pelcovits	
Thesis: “Geometry and Mechanics of Self-Assembling Colloidal Membranes”	
Sc.M. Engineering, Brown University	2017
Sc.M. Applied Mathematics, Brown University	2013
Certificates in Reflective Teaching, Course Design, and Teaching Consulting, Sheridan Center for Teaching and Learning, Brown University	2018
B.S. Physics, Georgia Institute of Technology	2012
B.S. Applied Mathematics, Georgia Institute of Technology	2012
Certificate in I/O Psychology, Georgia Institute of Technology	2012

Awards and Honors

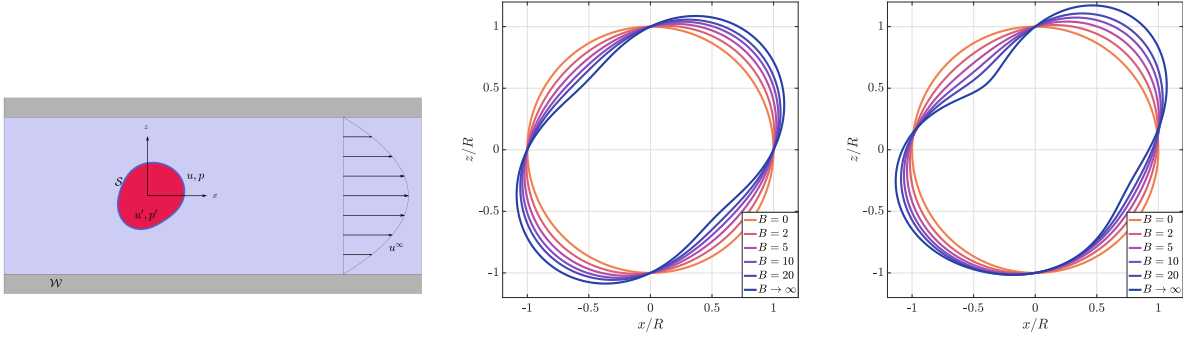
Sigma Xi, Brown University Chapter	2018
Outstanding Teaching Assistant, Brown University	2014
School of Mathematics MCTP Scholarship, Georgia Institute of Technology	2012, 2011
Hitohiro Fukuyo Memorial Award, Georgia Institute of Technology	2011

Research Interests

Applied mathematics: applied and numerical PDEs, differential geometry, data analysis
Soft matter & biophysics: membranes, fluid interfaces, colloids, liquid crystals, active matter
Biomechanical engineering: cellular and tissue mechanics, bioinstrumentation, biomaterials
Machine learning: metrology, inverse problems, data analytics, uncertainty quantification

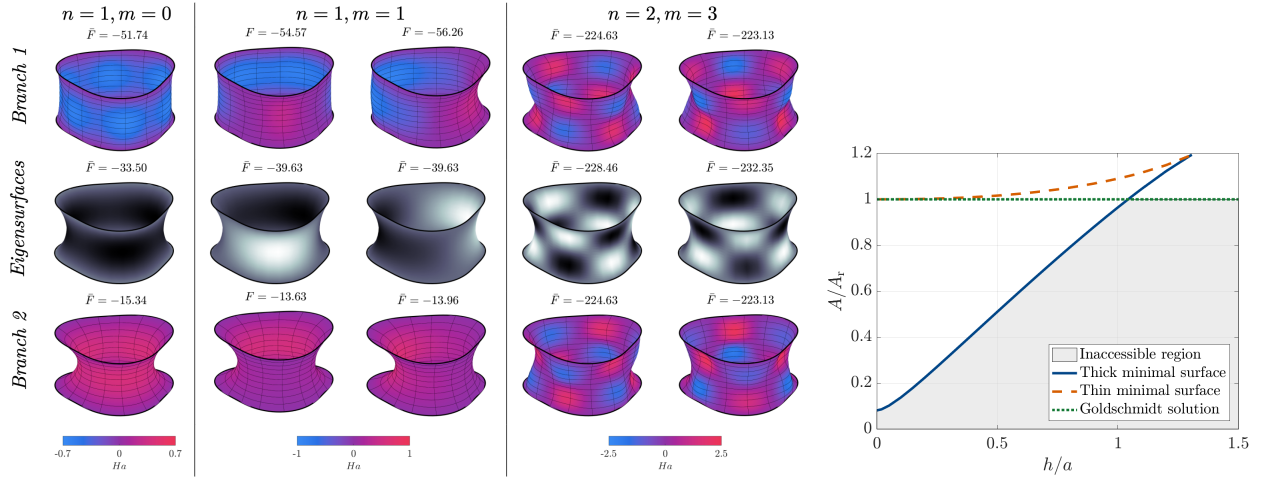
Publications

11. L. L. Jia. “The impact of interfacial elasticity on vesicle cross-stream migration.” To be submitted September 2026.



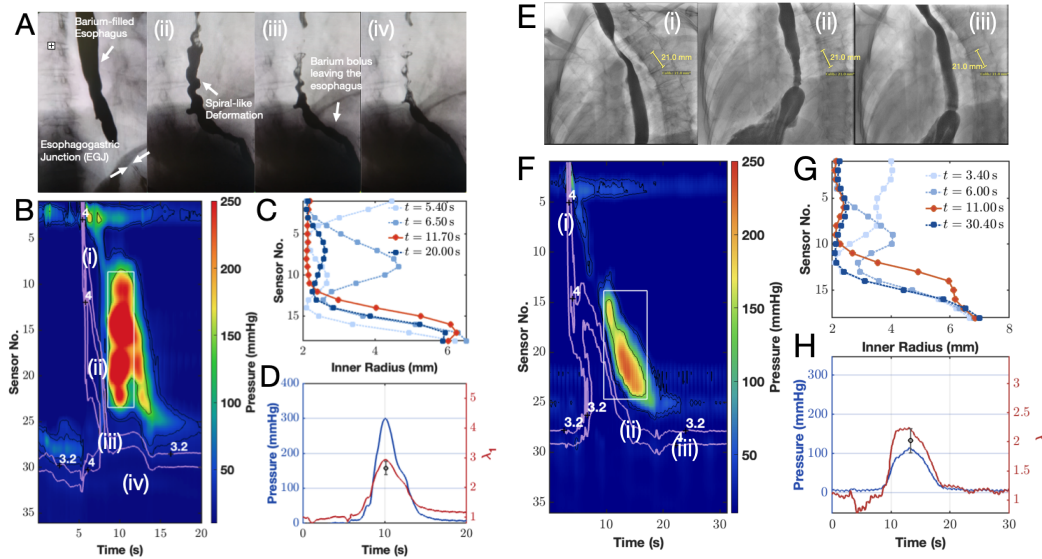
Abstract: Cells are frequently transported through confined fluid flows, and their ability to migrate across streamlines can strongly influence their biological function. We consider a simple extension of the model of Chan and Leal (J. Fluid Mech., vol. 92, 1979, pp. 131–170) for the migration of a fluid droplet in a parallel-plate channel that allows us to examine the steady-state motion of a deformable vesicle with a Helfrich bending resistance. The intimate mathematical connection between the two systems allows for the straightforward solution of what would normally be complicated nonlinear partial differential equations. The calculation of surface shape, particle velocity, and conditions leading to cross-stream migration are determined analytically for Poiseuille and shear flows, and some mistakes in the formulas of the original paper are pointed out. A brief comparison of our results with published data shows that the fundamental principles are captured in this calculation.

10. X. Liu, N. A. Patankar, L. L. Jia. “Shapes, forces, and torques of compressed elastic fluid interfaces: beyond axisymmetric configurations.” Under review (2026). arxiv: 2608.22567.



Abstract: Inspired by the classical Plateau–Douglas problem for soap films bounded by two closed curves, we solve an analogous problem for fluid interfaces with fixed surface area and resistance to out-of-plane bending. The boundaries are planar but need not be symmetric or concentric. The equilibrium surfaces minimize the Willmore bending energy and generalize minimal surfaces such as the catenoid while also exhibiting characteristic features of confined elastic interfaces such as buckling. We systematically classify all possible buckling modes and solution branches by performing a weakly nonlinear analysis and developing a fully nonlinear spectral solver. Our mathematical framework provides insight into the forces and torques required to stabilize cellular membranes and other soft materials and shows that physically relevant asymmetric states can arise even in symmetric systems.

9. X. Liu, L. L. Jia, M. M. Alikhan, W. Kou, D. A. Carlson, P. J. Kahrilas, J. E. Pandolfino, N. A. Patankar. “Mechanistic analysis of esophageal corkscrewing driven by active contraction.” Under review (2026).



Abstract: Corkscrew esophagus is a pathological condition in which the esophageal wall undergoes helical twisting or localized constrictions in response to strong muscular contractions. To unravel the mechanical origins of this phenomenon, we build a finite-deformation minimal model of the esophageal wall that incorporates an active thin-shell outer layer to capture the effect of contracting muscle fibers. We apply this model to measure muscle contractility in vivo using high-resolution manometry, and the corkscrew is predicted to emerge as an elastic instability when critical values of contractility are achieved. Analysis of the model and clinical data reveal plausible physiological changes in the esophageal muscle and matrix that precipitate instability including longitudinal muscle impairment and material softening due to repeated straining. Our framework links clinical observations to first-principles mechanics and provides quantitative targets for diagnostic thresholds and therapies.

8. G. R. Chickering, L. L. Jia, M. DiSalvo, M. A. Catterton, P. N. Patrone, E. M. Darling, G. A. Cooksey. “Estimating single-cell elastic modulus in a serial microfluidic cytometer from time-of-flight and fluorescence signals analysis.” *Lab on a Chip* **26** (2026) 3168.
7. R. Adkins, J. Robaszewski, S. Shin, F. Brauns, L. L. Jia, A. Khanra, P. Sharma, R. A. Pelcovits, T. R. Powers, Z. Dogic. “Topology and kinetic pathways of colloidosome assembly and disassembly.” *Proceedings of the National Academy of Sciences of the USA* **112**(36) (2025) e2427024122.
6. L. L. Jia, W. T. M. Irvine, M. J. Shelley. “Incompressible active phases at an interface. Part 1. Formulation and axisymmetric odd flows.” *Journal of Fluid Mechanics* **951** (2022) A36.
5. L. L. Jia, M. J. Shelley. “The role of monolayer viscosity in Langmuir film hole closure dynamics.” *Journal of Fluid Mechanics* **948** (2022) A1.
4. A. Khanra, L. L. Jia, N. P. Mitchell, A. Balchunas, R. A. Pelcovits, T. R. Powers, Z. Dogic, P. Sharma. “Controlling the shape and topology of two-component colloidal membranes.” *Proceedings of the National Academy of Sciences of the USA* **119**(32) (2022) e2204453119.

3. L. L. Jia, S. Pei, R. A. Pelcovits, T. R. Powers. “Axisymmetric membranes under external force: buckling, minimal surfaces, and tethers.” *Soft Matter* **17** (2021) 7268. Selected as Front Cover.
2. A. Balchunas, L. L. Jia, M. J. Zakhary, J. Robaszkowski, T. Gibaud, Z. Dogic, R. A. Pelcovits, T. R. Powers. “Force-induced formation of twisted chiral ribbons.” *Physical Review Letters* **125** (2020) 018002.
1. L. L. Jia, M. J. Zakhary, Z. Dogic, R. A. Pelcovits, T. R. Powers. “Chiral edge fluctuations of colloidal membranes.” *Physical Review E* **95** (2017) 060701(R).

In the News

1. L. L. Jia and X. Liu. “Mathematical Models Can Help Us Understand – and Possibly Treat – Complex Diseases.” NIST Taking Measure Blog (2025).

Teaching Experience

- Instructor, Momentum Scholars (August 2026–present). Design and teach physics, calculus, and engineering courses to high school students who do not have access to these courses at their schools.
- Lecturer, NIST Applied and Computational Mathematics Division Reading Group (January 2024–June 2024). Topics included the calculus of variations, differential geometry, and theoretical mechanics.
- Instructor, North Carolina School of Science and Mathematics (Spring 2023). Developed and taught a course on biomedical engineering.
- Instructor, Division of Applied Mathematics, Brown University (Summer 2017). Developed and taught a course on introductory scientific computing.
- Instructor and Peer Mentor, New Scientist Program Catalyst, Brown University (Summer 2014). NSP Catalyst was a program that mentored aspiring STEM majors with historically underrepresented identities. Developed and taught a course on complex numbers, basic differential equations, and linear stability theory.
- Teaching Consultant, Sheridan Center for Teaching and Learning, Brown University (Fall 2017–Spring 2018). Obtained Sheridan Center Reflective Teaching, Course Design, and Teaching Consultant Certificates.
- Graduate Teaching Assistant, Division of Applied Mathematics, Brown University (Fall 2013–Spring 2015). Taught methods of applied mathematics (differential equations). I received an “Outstanding TA” award in 2014.
- Undergraduate Teaching Assistant, School of Physics, Georgia Institute of Technology (Spring 2009–Spring 2012). Taught introductory mechanics and electromagnetism, 1 to 4 sections each semester.
- Undergraduate Teaching Assistant, School of Mathematics, Georgia Institute of Technology (Fall 2011–Spring 2012). Taught calculus and linear algebra, 1 section per semester.

- Undergraduate Grader, School of Mathematics, Georgia Institute of Technology (Fall 2010–Spring 2012). Graded for partial differential equations, abstract algebra, and real analysis courses.
- Math Resource Center Tutor, Brown University (Fall 2012–Spring 2018). Tutored undergraduate mathematics.
- Math Lab Tutor, Georgia Institute of Technology (Fall 2011–Spring 2012). Tutored undergraduate mathematics.
- I regularly work as a tutor (5-10 hours/week). I typically teach K–12 and college math and science as well as test preparation and math for competitions.

Students Trained

- Henry Nonnemaker, SURF student at NIST (Summer 2026). Henry is an undergraduate student majoring in Physics and English at Case Western Reserve University. He worked on computationally modeling high-strain-rate membrane expansion tests. He also published an article on his research for NIST’s official blog Taking Measure.
- Xinyi Liu, Guest Researcher at NIST (Spring 2024–Winter 2025). Co-advised with N. A. Patankar. Xinyi was a Ph.D. student in the Department of Engineering Sciences and Applied Mathematics at Northwestern University. She works on clinically inspired continuum mechanical models of the esophagus and membranes. She successfully defended her thesis in May 2026.
- Sydney Lee, SURF student at NIST (Summer 2024). Sydney is an undergraduate student at Texas A&M University. Her project involved calibrating and using a neuromorphic camera to quantify particle kinematics in microfluidic devices.
- Steven Pei, UTRA student at Brown University (Fall 2016–Fall 2018). Co-advised with T. R. Powers. Steven studied Physics and worked on theoretical and computational models of membranes, becoming a coauthor on a Soft Matter cover article. Upon graduation, he became a software engineer at Google.
- Xingjian “Jackson” Gao, high school student (Summer 2017–Fall 2017). Jackson was a student in my Scientific Computing course at Brown. I was Jackson’s senior project (he solved an optimization problem inspired by a video game in MATLAB) and college mentor. Jackson went on to study CS/applied math at UC Berkeley.
- College mentor for Chia-Yuan “Calvin” Chang, high school student (Summer 2017). Calvin went on to study CS/applied math at UC Davis.

Support

- My research has been supported by the NSF, the Simons Foundation, and NIST. I have written several successful proposals (and several unsuccessful proposals) for internal funding at NIST.
- I have also supervised multiple successful National Resource Council Postdoctoral Fellowship proposals.

- As a graduate student, I helped my advisors write a successful NSF proposal (NSF CMMI-1634552), which funded some of my work.

Presentations and Posters

- Interact talk + poster, APS Division of Fluid Dynamics Meeting (November 2026)
- Contributed talk, Modeling the Invisible Workshop (June 2026)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2025)
- Invited talk, SIAM/CAIMS Annual Meeting (July 2025)
- Poster, NIST ITL Science Day (December 2024)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2024)
- Poster, International Soft Matter Conference (July 2024)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2023)
- Seminar, National Institute of Standards and Technology (February 2023)
- Seminar, University of Alabama at Birmingham (February 2023)
- Seminar, University of Pennsylvania (January 2023)
- Seminar, National Security Agency (October 2022)
- Seminar, Flatiron Institute (October 2022)
- Seminar, Manhattan College (August 2022)
- Seminar, High Point University (July 2022)
- Seminar, Soft, Living, Active, and Adaptive Matter Seminar Series (June 2022)
- Poster, Mechanics of Life Workshop (May 2022)
- Seminar, The Cooper Union (April 2022)
- Seminar, Northwestern University (February 2022)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2021)
- Seminar, Flatiron Institute (September 2021)
- Contributed talk, APS March Meeting (March 2021)
- Seminar, Flatiron Center for Computational Biology Virtual Retreat (January 2021)
- Seminar, Flatiron Center for Computational Biology Brown Bag Seminar (May 2020)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2019)
- Contributed talk, APS March Meeting (March 2019)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2018)
- Poster, Brandeis University, MRSEC Site Visit (May 2018)
- Poster (w/ J. Robaszewski), Brandeis University, MRSEC Site Visit (May 2018)
- Poster (w/ A. Balchunas), Brandeis University, MRSEC Site Visit (May 2018)

- Invited talk, SIAM Conference on the Life Sciences (August 2018)
- Seminar, Flatiron Institute (May 2018)
- Seminar, Brown University (May 2018)
- Seminar, University of Massachusetts at Amherst (April 2018)
- Contributed talk, APS March Meeting (March 2018)
- Contributed talk, APS March Meeting (March 2017)
- Poster, Brown University, Department of Physics Poster Session (November 2016)
- Seminar, Brandeis University (October 2016)
- Contributed talk, New England Complex Fluids Workshop (September 2016)
- Contributed talk, APS March Meeting (March 2016)
- Seminar, Brown University Applied Math Graduate Seminar (March 2016)
- Presentation, Brown University Applied Math Graduate Retreat (September 2015)
- Seminar, Brandeis University (August 2015)
- Poster, UC Boulder, Boulder Summer School in Condensed Matter Physics (July 2015)
- Presentation, Kobe University (August 2014)

Service and Committees

- Doctoral thesis committee, Xinyi Liu, Northwestern University (Spring 2026)
- Co-chair (w/ Z. Grey), NIST Applied and Computational Mathematics Division Seminar Series (Summer 2024–Summer 2026)
- Events Committee, Association of NIST Asian–Pacific Americans (Fall 2023–Summer 2026) & Secretary, Association of NIST Asian–Pacific Americans (Fall 2024–Summer 2026)
- Georgia Tech Alumni Association Mentor (Fall 2024–present)
- Reviewer, NIST Small Business Innovation Research grant proposals (Spring 2024)
- Member, Simons Foundation AAPI Employee Resource Group (Winter 2021–Fall 2022)
- Professional Development Committee, Flatiron Institute (Winter 2021–Fall 2022)
- Referee, Journal of Fluid Mechanics ($\times 5$)
- Brown University New Teaching Assistant Orientation Facilitator (Fall 2017)
- Brown University Applied Math Grad-Undergrad Mentor (Spring 2016–Spring 2018)
- Brown University New International Graduate Student Orientation Peer Mentor (Fall 2016, Fall 2017)
- Brown Applied Math Graduate Retreat Project Leader (Fall 2016)

Personal

- Languages: English (native), Mandarin Chinese (fluent), Japanese (conversational)
- Memberships: APS (DFD, DSOFT, DBIO), SIAM

References

Ms. Nikki Geula (teaching reference)
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Dept. of Physics & Dept. of Biomolecular Science and Engineering, UCSB
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Prof. Michael Shelley (postdoctoral supervisor)
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